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He, Xinxin; Li, Zhiyuan; Huang, Xingcan; Zhang, Qiang; Zeng, Yuyang; Li, Jialin; Yiu, Chun Ki; Yang, Yawen; Zhou, Jingkun; Xu, Guoqiang; Wang, Jiachen; Li, Jian; Xu, Zitong; Chen, Zhenlin; Liu, Yiming; Gao, Yuyu; Zhang, Binbin; Zhao, Guangyao; Gao, Zhan; Wu, Pengcheng; Shi, Rui; Qiu, Yuze; Zhang, Hehua; Chow, Lung; Ye, Denglin; Huang, Ya; Yu, Xinge

Published in:
BMEMat

Published: 01/03/2025

Document Version:

Final Published version, also known as Publisher's PDF, Publisher's Final version or Version of Record

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Publication record in CityUHK Scholars:

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Published version (DOI):

[10.1002/bmm2.12124](https://doi.org/10.1002/bmm2.12124)

Publication details:

He, X., Li, Z., Huang, X., Zhang, Q., Zeng, Y., Li, J., Yiu, C. K., Yang, Y., Zhou, J., Xu, G., Wang, J., Li, J., Xu, Z., Chen, Z., Liu, Y., Gao, Y., Zhang, B., Zhao, G., Gao, Z., ... Yu, X. (2025). Sweat-powered, skin-adhesive multimodal sensor for long-term and real-time sweat monitoring. *BMEMat*, 3(1), Article e12124.
<https://doi.org/10.1002/bmm2.12124>

Citing this paper

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RESEARCH ARTICLE

Sweat-powered, skin-adhesive multimodal sensor for long-term and real-time sweat monitoring

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Funding information

Research Grants Council, University
 Grants Committee, Grant/Award
 Numbers: 11211523, 11213721, 11215722,
 RFS2324-1S03; National Natural Science
 Foundation of China, Grant/Award
 Number: 62122002; City University of
 Hong Kong, Grant/Award Numbers:
 9667199, 9667221, 9667246, 9680322

Abstract

The importance of continuous healthcare management has significantly accelerated the development of wearable devices for monitoring health-related physical and biochemical markers. Despite extensive research on wearable devices for physiological and biochemical monitoring, critical issues of power management and device/skin interfacial properties restrict the advancement of personalized healthcare and early disease detection. Here, we report a multimodal sweat monitoring device featuring a real-time display and long-term data analysis based on self-powered format of sweat-activated batteries (SABs). The polyvinyl alcohol-sucrose (PVA-Suc) hydrogel serves as the key component for the SAB, offering not only great long-term adhesive properties for conformable wearability but also significant power generation capabilities. A maximum current density of 44.06 mA cm⁻² and a maximum power density of 21.89 mW cm⁻² can be realized for the hydrogel based SAB. The resulting device integrates an advanced colorimetric and electrochemical sensor array to measure pH levels, glucose concentrations, and chloride ion levels in human sweat, with data wirelessly transmitted by Near Field Communication. The self-powering features and multiple mode sensing function offer sufficient power to support real-time monitoring of metabolic biomarkers in sweat, with the ability to visually observe changes in the colorimetric sensors for long-term data monitoring.

KEYWORDS

adhesive hydrogel, healthcare management, sweat monitoring, sweat-activated batteries, wearable sensors

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1 | INTRODUCTION

Wearable devices have been developing rapidly in response to the greatly increased interest in healthcare management, digitally assisted fitness, and physical condition monitoring.^[1,2] Over the past decade, substantial research has been devoted to wearable sensors,^[3–6] including advances in functional materials,^[7] clinically driven new biomarkers,^[8] and the development of wearable devices with excellent biocompatibility and mechanical properties.^[9–13] Wearable biosensors, offering fast and efficient real-time non-invasive detection, have made great contributions to monitoring and measuring human physiological state, enabling non-invasive health monitoring for sport activities, personalized nutrition, and high-precision medicine.^[14–18] As health-related information can be continuously monitored with on-body sensors, these data can supplement laboratory- or medical institution-based testing.^[19] Among the existing wearable sensing modalities, biomarkers are among the most important as they are closely related to specific diseases and can aid in the prevention and early diagnosis.^[20–24] Sweat, one of the most common and easily collected body fluids, has received special attention in the field of personalized medical monitoring devices due to the huge potential for non-invasive collection of important biomarkers, such as various ions, glucose, lactic acid, etc.^[25] whose concentrations play an important role in indicating health status.^[26–28]

Long-term and stable sweat monitoring has spurred the development of numerous wearable biosensors.^[29] However, these systems still face challenges in energy provision and stable skin/device interfaces.^[30] Conventional batteries compromise portability, flexibility, and biocompatibility due to their rigidity and the potential for electrolyte leakage and contamination.^[26,29] Additionally, most reported sweat monitoring systems rely on Bluetooth for data transmission and reading, which significantly increases energy demands due to the frequent reading steps.^[31] In recent years, the advancement of sweat-activated batteries (SABs) has provided a promising solution powered by the electrochemical reactions between sweat and electrodes.^[32–34] Due to the continuous secretion of sweat, a sustained power supply can be achieved for long-term data recording and transmission in sweat sensors.^[35] Furthermore, SABs can be designed in ultrathin, flexible, and stretchable formats that conform to various complex surfaces.^[36] And the materials used in SABs are also biocompatible and replaceable, ensuring safety and long-term power generation.^[31,37,38] Another challenge in long-term and stable sweat monitoring is interfacial stability between

wearable sensors and skin, resulting from insufficient adhesion between the sensors and skin.^[39–42] Most adhesive polymers, such as commonly used Polydimethylsiloxane (PDMS) in wearable electronics, often easily lose adhesion due to the formation of gaps and interfacial water between the skin and the device caused by sweat secretion.^[43] Such issues may lead to allergic reactions or detachment of the device during physical activities.^[44] Therefore, the question arises whether it is possible to develop functional adhesive materials and devices to ensure both high power output from the SAB and good interfacial adhesion properties with skin that can be used for long-term and continuous sweat monitoring in a self-powered format.

Here, we report a skin-adhesion, multimodal, sweat-activated system (SAMS) which incorporates SABs specifically designed to power optimized wearable sweat sensors, and thus provide long-term stable sweat monitoring for health status detection. To further increase the operational period of the SABs, we integrated a colorimetric module and an electrochemical sensing module within one single system, measuring key biomarkers such as glucose concentration, pH levels, and chloride ion concentrations in human sweat. The colorimetric sensors provide an immediate response to sweat concentration without power consumption.^[45,46] At the same time, SABs exclusively supply power to the electrochemical sensing module with data stored in a memory chip. The integration reduces the need for frequent data transmission between sensors and external equipment while also allowing timely data reading. This design enhances the operational efficiency and extends the usage duration of the SABs, significantly improving the overall performance of the system. A key technique for the integration is the use of adhesive and self-healing polyvinyl alcohol-sucrose (PVA-Suc) hydrogel. The electrolyte of SABs is achieved by adding salt to the PVA-Suc hydrogel without significantly compromising adhesion. The self-healing property ensures the cohesive integration of SABs, colorimetric sensors and the adhesive layer of the device, facilitating the replacement of color-changed colorimetric sensors. The PVA-Suc hydrogel maintains good adhesion even after absorbing large volumes of sweat, ensuring a stable interface between device and skin. Finally, users can read the sweat signals in real-time on the colorimetric sensors and analyze long-term signals when they access the device via a smartphone, with data transmitted by a Near Field Communication (NFC) system. This system ensures that users have immediate and easy access to their health metrics in a user-friendly format and also ensures accurate and long-term sweat monitoring.

2 | RESULTS AND DISCUSSION

2.1 | Sweat-activated, skin-adhesive multimodal system

A schematic illustration of the SAMS designed for comprehensive health monitoring through sweat analysis is shown in Figure 1a. The system is composed of multiple layered components that work together to monitor the information of the three sweat components. The uppermost layer consists of a Flexible Printed Circuit Board (FPCB) featuring the electrical circuit and electrical elements on the front, along with NFC coils and four sensing electrodes on the reverse side, one of which serves as a reference electrode. Directly beneath this is a compact flexible SAB that powers the system upon activation by sweat. It supplies energy sufficient for the microcontroller unit (MCU) to perform data collection and storage functions in memory. The base layer includes a PVA-Suc hydrogel which functions both as an adhesive and a colorimetric sensor. This hydrogel is notable for its robust adhesion and self-healing properties, allowing for seamless integration or replacement with other colorimetric sensors, thus enhancing the adaptability and efficiency of the system in health monitoring applications. Figure 1b depicts a top-view schematic diagram of the SAMS highlighting the internal structure of the PVA-Suc hydrogel within the colorimetric sensor. The hydrogel is thermally crosslinked from polyvinyl alcohol (PVA) and sucrose, creating numerous active hydrogen bonds that enhance its adhesion, mechanical properties, and moisture retention. These properties ensure seamless contact with various surfaces, which is crucial for accurate sensor performance. In general, the materials required for PVA-Suc hydrogels are straightforward, and the preparation method is simple. Upon the addition of NaCl, the resulting PVA-Suc-NaCl serves as an effective conductive medium in SAB. Moreover, due to its self-healing properties, it integrates effectively with the adhesive layer, forming a cohesive unit. As shown in Figure 1c, the SAB is located at the sides of the circuit structure and consists of an anode of magnesium (Mg) foil, a cathode of nickel/graphene/MnO₂ foam, linked by PVA-Suc-NaCl hydrogel with superior air-permeability and catalysis ability for oxygen reduction, as well as an electrolyte bridge. When the hydrogel was rehydrated by absorbing sweat, the SAB would be activated and generate sufficient power to operate the MCU and memory at the same time, achieving real-time data acquisition, processing, storage, and multiplexed monitoring of key biomarkers in sweat.

The wearable devices used for sweat monitoring need to continuously and stably detect the user's physiological indicators, so the encapsulation materials need to have

good flexibility and mechanical stability. However, the mismatch between the common packaging materials used for flexible sensors may lead to sweat accumulation, making the test results inaccurate. As illustrated in Figure 1d-ii, the SAMS shows superior skin adhesion with the rapid sweat absorption of PVA-Suc hydrogels, ensuring a secure fit to the skin. In contrast, other encapsulation materials, such as PDMS, are nearly hydrophobic and create gaps that can lead to a loose fit and diminished sensor effectiveness, as demonstrated in Figure 1d-i.^[47–49] Furthermore, the self-healing properties of the PVA-Suc hydrogel are showcased in Figure 1d-iii. Initially intact, the hydrogel, when cut and pressed together for about 1 minute, heals effectively, nearly erasing the cut line. This property enhances the durability of hydrogels making it particularly useful for colorimetric sensors where stability and longevity are crucial. Meanwhile, the property of self-healing can also make the preplacement of SAB more convenient.

As shown in Figure 1a,e flexible dual-mode monitoring platform was powered by SABs. Benefitting from the advanced power sources and integration strategies, this system is able to continuously monitor three physiological signals in human sweat-glucose, pH, and chloride ion levels, both colorimetric and electrical sensors in situ. The colorimetric sensors are applied for the direct observation of the real-time health status, while the electrical sensors are applied to precisely monitor and record the sweat information combined with the SAB-powered MCU and memory unit. When the subject begins to sweat, the sweat will quickly activate the SAB, providing a constant voltage for the SAMS. The SAB power supply enables the MCU and memory unit to work, the MCU processes the data collected by the electrical signal sensor, and the memory unit stores the voltage difference signals for data transmission. Data from the sensors are transmitted to a mobile device via an NFC connection, providing users with detailed and continuous data for over 60 min. This integrated approach offers precise and comprehensive monitoring, enhancing user health management insights. Finally, compared with five other sweat-related battery technologies, the SAMS shows excellent properties in the realm of compact size, high current density, strong adhesion, superior power density, colorimetric sensor, and dual mode (Figure 1f).^[37,38,50,51]

2.2 | Performance of the PVA-Suc hydrogel

Sucrose enhances the porosity of PVA hydrogels upon water absorption, as depicted in Figure S1 of the

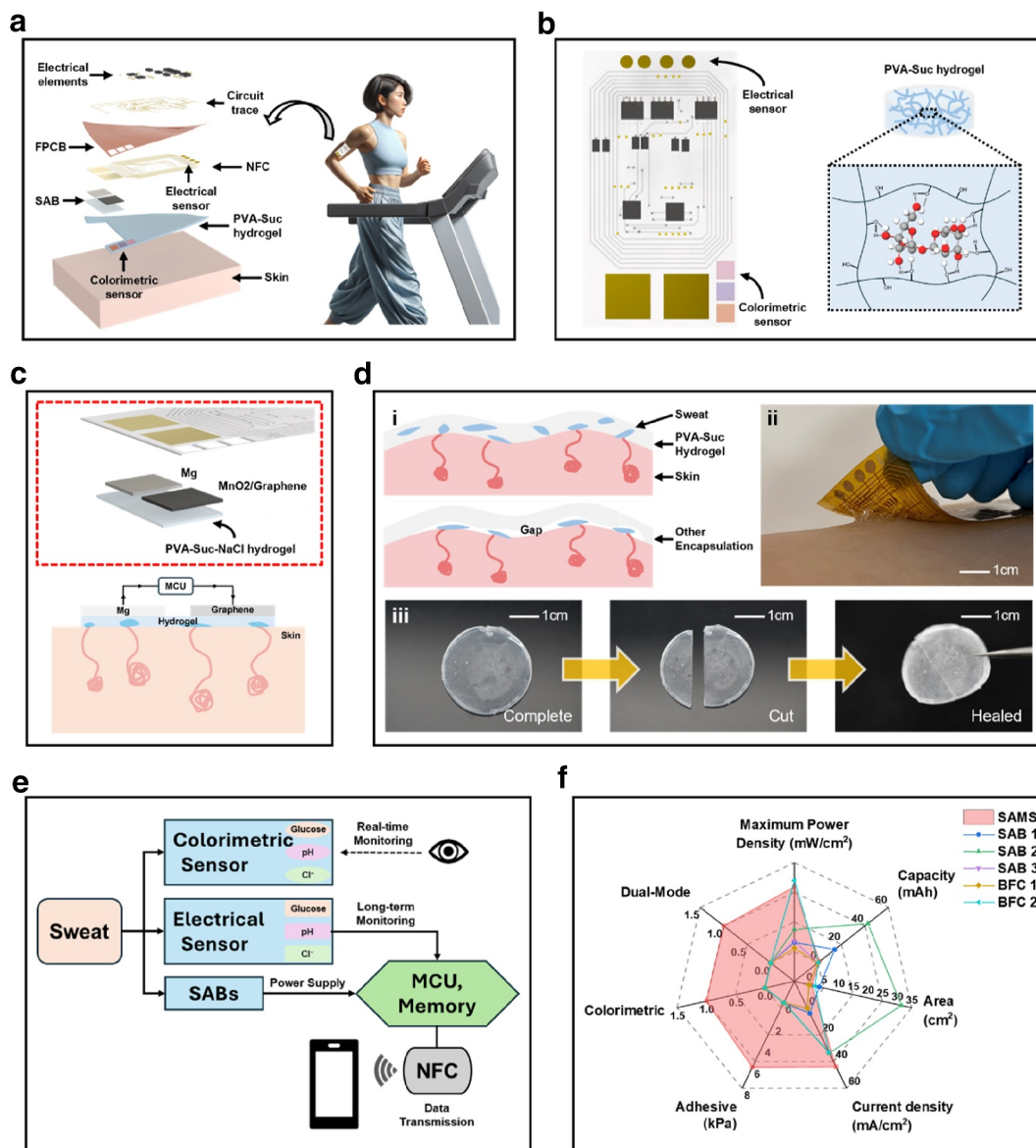


FIGURE 1 Schematic diagram and optical images of the SAB-powered dual-mode sweat monitoring system. (a) Schematics of the SAMS wearable platform for people in motion. (b) Schematic diagram of the SAMS and PVA/Suc hydrogel. (c) Schematic exploded view of SABs (up). Schematic diagram of the working principle of the SABs (down). (d) Schematic diagram of PVA-Suc hydrogel adhesion. (i) Schematic diagram of water absorption of PVA-Suc hydrogel, while other packaging materials may have interfacial water between them and the skin. (ii) Optical image of the adhesive hydrogel. (iii) Self-healing process of hydrogels. (e) Block diagram of the design for the sweat-activated monitoring patch. (f) Battery performance comparison diagram among SAMS, SAB and Bio-fuel cells (BFC). BFC, Bio-fuel cells; PVA/Suc, polyvinyl alcohol-sucrose; SAB, sweat-activated batteries; SAMS, sweat-activated system.

supporting information. Furthermore, the improved mechanical properties of the proposed PVA-Suc hydrogels ensure structural integrity under various conditions, thereby broadening their potential applications, and the superior water retention capability of PVA-Suc hydrogel not only prolongs its functional lifespan but also stabilizes sensor performance by maintaining optimal hydration levels.^[52] We investigated the chemical structure of the PVA-Suc composite hydrogel using Fourier-transform infrared spectroscopy (FTIR), shown in Figure 2a. The

broad absorption bands around 3275 cm^{-1} , which are indicative of the stretching vibrations of hydroxyl groups, are more pronounced in the PVA-Suc composite than in pure PVA or sucrose alone. This suggests stronger intermolecular hydrogen-bonding interactions within the composite. The peak at 2928 cm^{-1} is attributed to asymmetric C-H stretching vibrations, and the peaks at 1417 , 1329 , and 984 cm^{-1} correspond to C-O-H bending vibrations of sucrose. The band at 1040 cm^{-1} is related to the C-O stretching vibrations, while the bands at 920 and

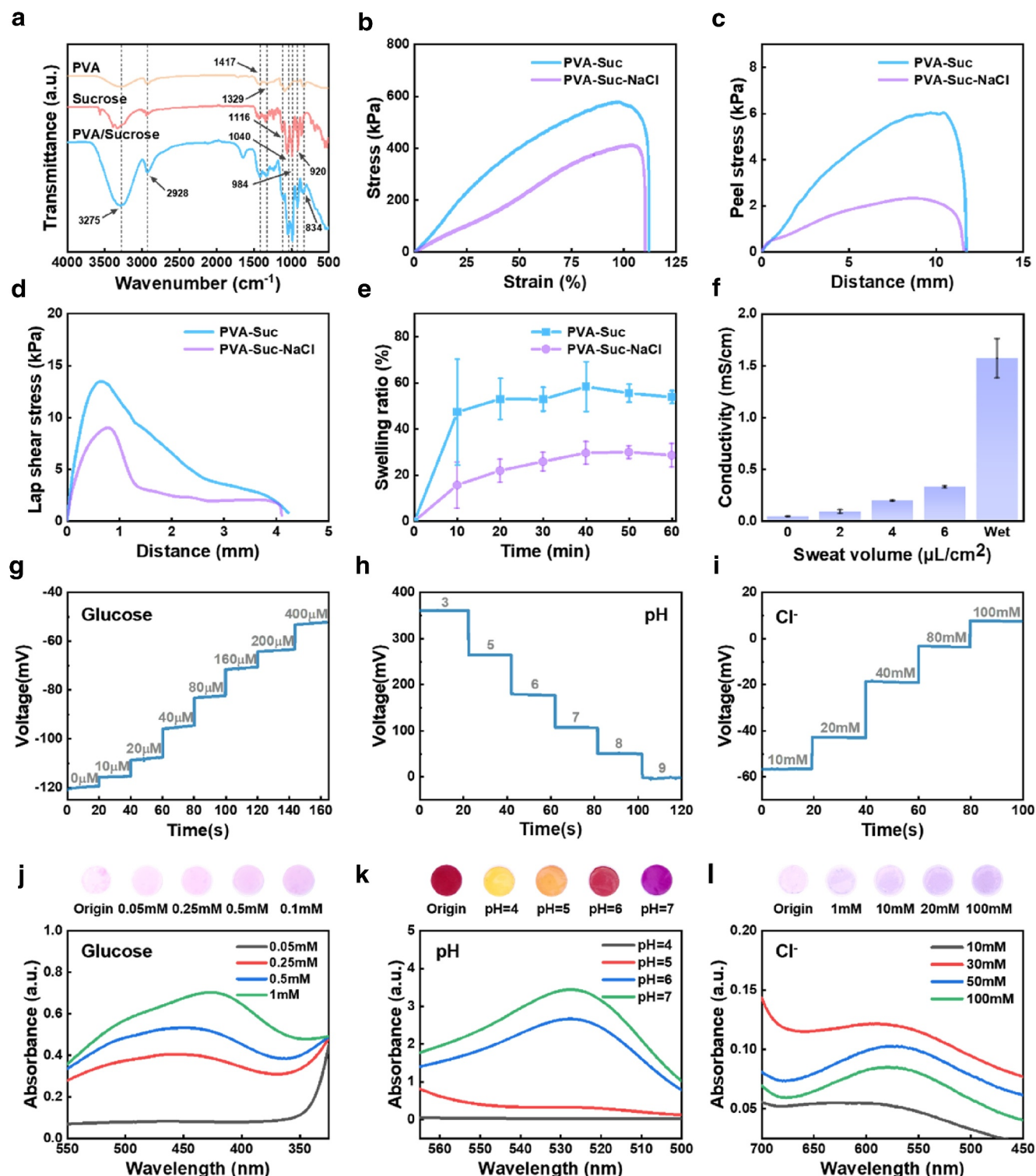


FIGURE 2 Characteristics of the hydrogel, electrical sensor and colorimetric sensor. (a) FTIR spectra: characteristic peaks. (b) Tensile test: stress-strain curves. (c) Peel test: peel stress-displacement curves. (d) Lap shear test: shear stress-displacement curves. (e) Swelling test: swelling ratio-time curves. (f) Conductivity of SABs in different sweat volumes. (g) Open circuit voltage of the glucose sensor as a function of time with increasing glucose concentration. (h) Open circuit voltage of the pH sensor as a function of time with increasing pH value. (i) Open circuit voltage of the chloride sensor as a function of time with increasing chloride concentration. (j) Optical images and UV-vis spectra of the glucose colorimetric sensor with different glucose concentrations. (k) Optical images and UV-vis spectra of the pH colorimetric sensor with different pH values. (l) Optical images and UV-vis spectra of the chloride colorimetric sensor with different chloride concentrations. FTIR, Fourier-transform infrared spectroscopy.

834 cm^{-1} are linked to the C-C stretching vibrations. Notably, the ring asymmetric valence vibration at 1116 cm^{-1} , present only in the sucrose and PVA-Suc spectra, confirms the successful integration of sucrose into the hydrogel matrix.

Meanwhile, dynamic mechanical testing was carried out to evaluate the stability and flexibility of the hydrogels. As shown in Figure 2b, the tensile stress-strain curves display a fracture strength of approximately 579.21 kPa for PVA-Suc, while PVA-Suc-NaCl exhibits a reduced strength of about 393.93 kPa due to the addition of electrolyte. Both hydrogels demonstrated significant elongation of around 110%. In Figure 2c, the peel stress-displacement curve is shown, indicating that higher peel stress is exhibited by PVA-Suc (~6.04 kPa) compared to that exhibited by PVA-Suc-NaCl (~2.33 kPa). Furthermore, we explored the skin adhesion of PVA-Suc hydrogel. As illustrated in Figure 2d, the shear stress-displacement curves, with PVA-Suc achieving a maximum lap shear stress of ~15.49 kPa, significantly higher than that of PVA-Suc-NaCl (~5.06 kPa). Also, their swelling ratio curves are studied in Figure 2e, noting that the addition of NaCl enhances electrical conductivity but slightly impairs mechanical integrity due to the disruption of hydrogen bonds. The data in Figure 2f indicate that an open-circuit voltage of 0.79 V is yielded by the activation of the sweat battery with minimal sweat accumulation (1 $\mu\text{L cm}^{-2}$), which underscores the efficiency of the SAB in generating a substantial voltage output from very low volumes of sweat, highlighting its potential applicability in low-exertion scenarios where minimal perspiration is expected.

The potentiometric glucose sensors use glucose oxidase (GOx) as the sensing component. For glucose sensor fabrication, the bottom layer is spray-coated with zinc oxide (ZnO), whose tetrapartite structure offers significant advantages in catalytic reactions and signal transmission,^[53–55] and chitosan is applied on the top layer to immobilize GOx. As shown in Figure 2g, the voltage response of the glucose electrical signal sensor to different glucose concentrations (0–400 μM) is illustrated. Similarly, pH sensors are prepared using a potential sensing method, where polyaniline (PANI) serves as the primary sensing component, known for its high response, reversibility, and specificity to pH changes,^[56] and the voltage response of the pH electrical signal sensor to different pH values (3–9) is shown in Figure 2h. The pH range of human sweat varies from 4.2 to 8.3.^[57,58] The pH electrical signal sensor in SAMS can accurately encompass the entire pH spectrum necessary for detection, and consistently exhibits a linear response between pH 3 and pH 9 with the R^2 of 0.99. For the detection of Cl^- , a layer of poly(3,4-ethylenedioxythiophene): sodium polystyrene

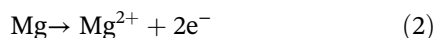
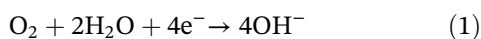
sulfonate (PEDOT: PSS) is deposited on the electrode to minimize potential drift and enhance electrode performance.^[6,59] A Cl^- selective membrane is then applied to the surface of PEDOT: PSS for Cl^- selective electrode fabrication,^[60] and the voltage responses of the Cl^- electrical signal sensor to different Cl^- concentrations (10–100 mM) are illustrated in Figure 2i. Figures S2 and S3 confirm that all sensors exhibit linear performance and robust anti-interference properties across relevant concentration ranges.^[61–63] As demonstrated in Figure S2a, the calibration curve depicting voltage versus glucose concentration exhibits perfect linearity, with a coefficient of determination $R^2 = 0.99$ and a slope of 38.581. This result aligns with the anticipated linear increase in open circuit voltage proportional to rising glucose concentrations.^[64,65] A calibration curve of voltage versus pH level is shown in Figure S2b; the curve is linear, reflected by an $R^2 = 0.99$. The slope of the fitted curve is -60.65 , closely aligning with the theoretical value of 59 mV/pH derived from the Nernst Equation. It indicates that the fabricated pH sensors demonstrate near-Nernstian behavior. The calibration curve of voltage versus chloride ion concentration is linear, as indicated by $R^2 = 0.99$ and depicted in Figure S2c. The slope of the fitted curve, measured at 65.53, is close to the theoretical value expected for chloride response.

The optical images, and UV-vis absorption spectra of the sensors are displayed in Figure 2j–2l, showing visible color changes and spectral shifts corresponding to varying concentrations of glucose, pH, and Cl^- , demonstrating their sensitivity and dynamic response characteristics.

2.3 | Design and performance of the SABs

The SAB represents a breakthrough in dual-mode wireless monitoring patches, epitomizing innovation in sustainable energy technologies. This device stands out for its compact and flexible design, making it ideal for integration into wearable electronics. The environmentally conscious design of the SAB underscores its potential to transform the bio-powered electronics sector by merging biotechnology with energy engineering, thereby expanding the horizons for personal and medical technologies.^[66] The functional components of the SAB include a nickel foam cathode coated with graphene and manganese dioxide (MnO_2), a magnesium foil anode, and a hydrogel electrolyte composed of polyvinyl alcohol (PVA), sucrose, and sodium chloride (NaCl). MnO_2 is critical as it enhances the oxygen reduction reaction (ORR) at the cathode, thus boosting the power output

and ensuring stable operation. The SAB operates as a magnesium-air battery, activated by the sweat absorbed into the hydrogel, which dissolves NaCl and establishes an ionic pathway between the anode and cathode. As sweat permeates the hydrogel, Na⁺ ions migrate towards the cathode, initiating the reduction reaction (Equation 1), while at the anode, magnesium oxidation releases Mg²⁺ ions and electrons (Equation 2). This ionic movement maintains electrical neutrality and facilitates the simultaneous oxidation of magnesium and the reduction of oxygen, catalyzed by MnO₂, to generate electrical energy.^[67]



The inclusion of graphene in the cathode can increase both the surface area and conductivity, allowing for more efficient electron transfer and higher power output. The design and functionality of the SAB mark significant progress at the intersection of biotechnology and energy engineering, promising scalable, cost-effective production and biocompatible energy solutions for wearable devices.^[68] In the SAMS model, the SAB uses a carbon cloth coated with MnO₂ as the cathode to facilitate the ORR, while a 10 mm by 10 mm Mg foil serves as the anode. The integration of MnO₂ on the cathode has been shown to quadruple the current output due to its superior catalytic performance in the ORR (Figure 3a).^[69] The spatial arrangement of the electrodes within the SAB is crucial for defining the device's dimensions, longevity, and efficiency. As demonstrated in Figure 3b, reducing the distance between electrodes minimizes impedance and enhances the short-circuit current. Specifically, the maximum short-circuit current decreases from 4.05 to 1.62 mA/cm² as the electrode separation increases from 1 to 3 and to 5 mm, while maintaining a magnesium foil thickness of 0.1 mm caused by the increase of the inner impedance. Furthermore, the thickness of the magnesium foil anode significantly influences the operational duration and performance of the battery. Increasing the anode thickness extends the operational time of the SAB from approximately 1.5 h to over 5 h, as shown in Figure 3c. The flexibility of the Mg foil decreased with increasing thickness from 0.1 to 0.3 mm, which affected the wearing comfort of SAMS, shown in Figure S8 in the supporting information. As a result, the Mg foil with a 0.1 mm thickness would be the optimal selection for balancing conductivity and flexibility. As illustrated in Figure 3d, the optimized SAB achieves a maximum current density of 44.06 mA/cm² and a maximum power

density of 21.89 mW/cm², sufficient for powering the device. Stability tests, as presented in Figure 3e, indicate that the open-circuit voltage of SAB remains stable between 1.51 and 2.13 V for over 2 h, underscoring its stability and reliability, and verifies the battery's ability to maintain a stable output over extended periods, which is essential for devices used in health monitoring that require consistent data collection and processing. Additionally, as shown in Figure 3f, the SAB's discharge capacity reaches 1.6 mAh when the output voltage drops from 1.7 to 0 V, which provides the SAMS with power for over 90 min. The relationship between the volume of artificial sweat and the electrical performance of the SAB was also studied, revealing that the open circuit voltage increases with the amount of sweat introduced (Figure 3g). The SAB requires only 2 µL/cm² of sweat to generate the necessary voltage to power the SAMS, facilitating rapid activation during exercise. It supplies energy to the MCU and the memory unit for processing and storing data but not to the electrical sensor. Therefore, variations in voltage do not compromise the accuracy of our detection system. Consequently, the continuous and stable operation of the SAMS does not depend on the SAB achieving its maximum output voltage. Bending and twisting tests evaluate the ability of SAB to endure physical deformations, which are common in daily wear, ensuring the battery maintains its structural integrity and performance consistency under stress. As shown in Figure 3h,i, the small voltage fluctuation and rapid recovery of the proposed SAB during physical deformations ensure that the device performs reliably over its lifespan.

2.4 | On-body real-time monitoring

The optical image of SAMS is shown in Figure S4 with the configuration of an FPCB layer, an SAB section on one side of the system, and a layer of PVA-Suc hydrogel. As illustrated in Figure S7 of the supporting information, the front side of the FPCB integrates the signal path and electronic components, and the back side is printed with NFC coils and four electrodes, which are three electrical signal sensors and a reference electrode. Three colorimetric sensors are located next to the SAB, along with electrical signal sensors, for real-time and long-term monitoring of glucose and chloride ion concentrations and pH values in sweat, respectively. To ensure adhesion and biocompatibility in contact with the skin, other areas on the back of the system are coated with PVA-Suc hydrogel, which not only improves the comfort and wearability of the device but also protects the electronic

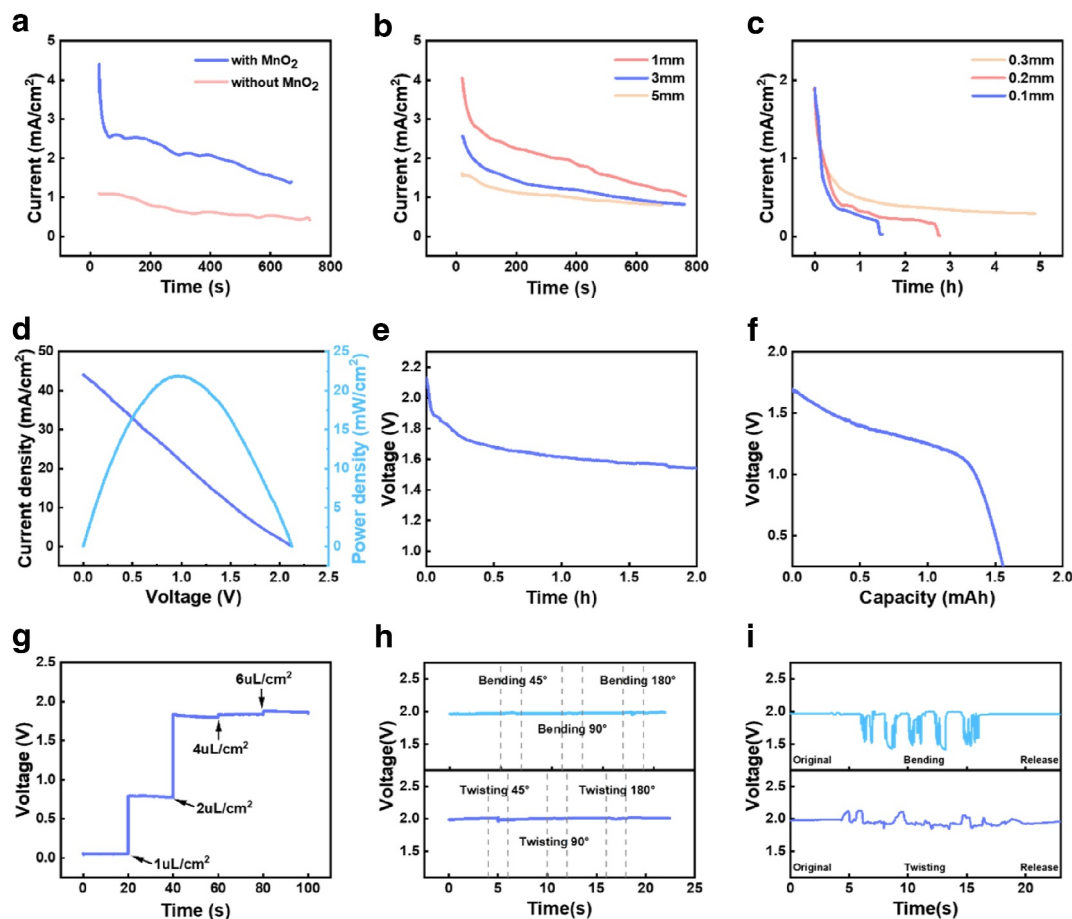


FIGURE 3 Performance characterization of the SABs. (a) Comparison of SABs when the anode with or without MnO_2 . (b) Short-circuit current of SABs with different electrode distances. (c) Short-circuit current of the SABs with three different thicknesses of Mg foil. (d) Current density and power density curve of the SABs. (e) Open-circuit of SABs as a function of time. (f) Capacity of SABs. (g) Electrical response of the SABs as a function of time with increasing injected sweat volume. (h) Electrical output response of the SABs bent to 45° , 90° , and 180° as a function of time (up). Electrical response of the SABs as a function of time at three different twisting angles ranging from 45° to 180° (down). (i) Electrical output response of the SABs under continuous bending and twisting. SABs, sweat-activated batteries.

components from environmental factors such as sweat and mechanical stress.

As demonstrated in Figure 4a, the SAMS can be applied to various body areas-forearms, chest, and back-as well as the flexibility and bendability, which can tailor to specific monitoring needs. The use of sweat produced during physical activity as a stimulus to activate the SAB system makes it one of the solutions for the long-term stable work of wearable sweat monitoring devices. In a practical application, as shown in Figure 4b-i, SAMS attaches to the arm of a subject, collects information on changes in glucose concentration, pH, and chloride ion concentration in sweat, and transmits signals via NFC. As demonstrated in Figure 4b-ii,c, two microcontrollers manage and refine the signals, with TSV632IYDT amplifiers boosting the signal quality. The sensor patch converts the collected data into digital format, stores it via I2C, and transmits it to a mobile device through an NFC chip. The mobile graphical user

interface of SAMS displays and logs the data for further analysis. The SAB powers the entire system, which is compact and delivers high current output, offering a lightweight (approximately 1.5 g) and smaller alternative to conventional Bluetooth battery systems. Furthermore, the electronic components were designed to minimize interference from external noise, enhancing the robustness of the system and ensuring high-quality NFC communication. As illustrated in Figure 4b-iii, after the start of sweating, the colorimetric sensor responds three times at intervals (0, 10, and 20 min) during exercise. The pH value gradually changes from purple to orange with the increase of exercise time, indicating a gradual decline in pH value, which is consistent with the electrical signal data in Figure 4f. The color of the glucose colorimetric sensor was slightly pink and remained constant, revealing the maximum glucose concentration in the sweat of the subject during exercise. Also, the color of the chloride ion

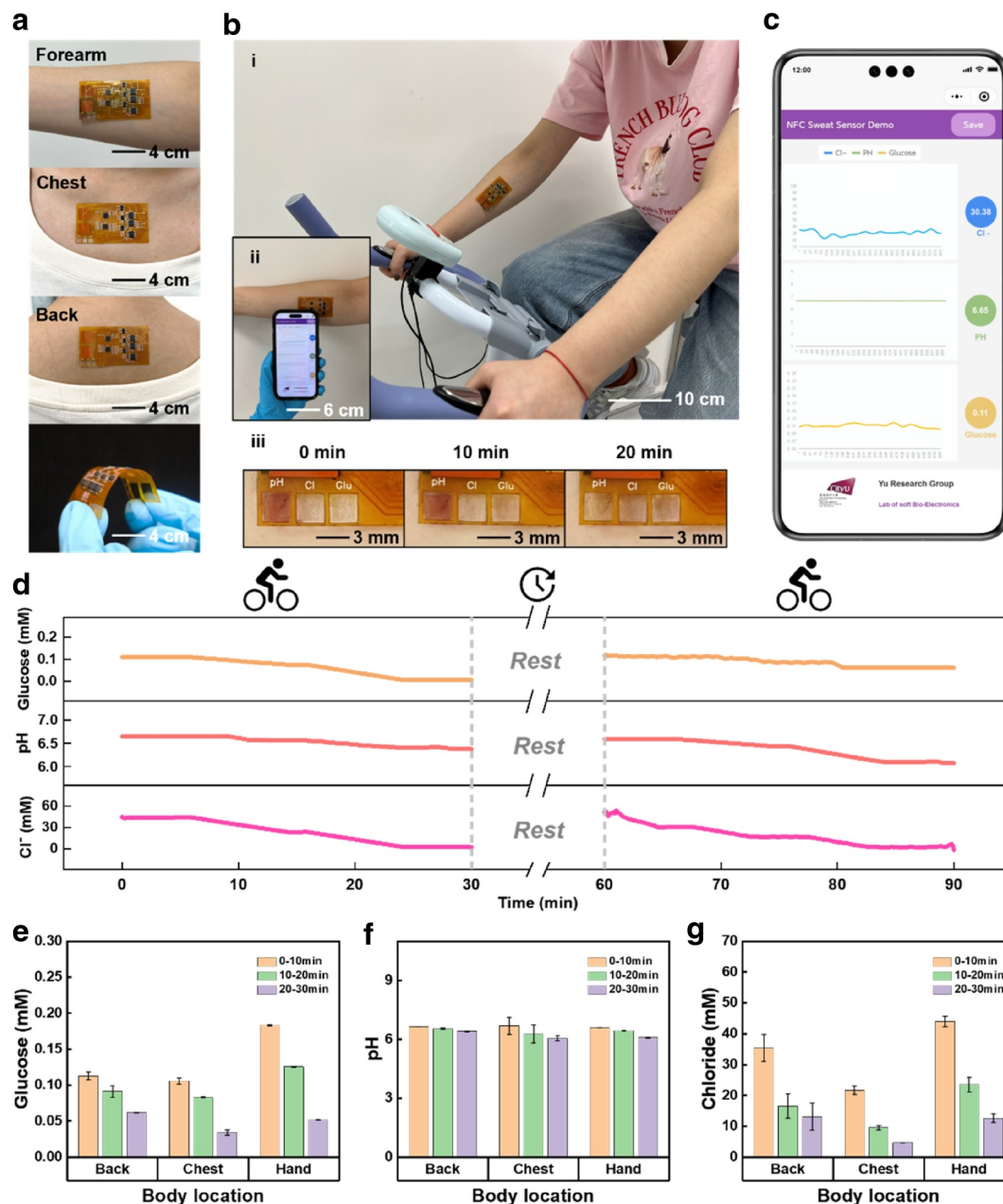


FIGURE 4 On-body sweat monitoring and wireless data transmission performance. (a) Optical image of the device attached to different body locations and the flexibility. (b) (i) The physical image of subjects wearing SAMS during exercise. (ii) Obtain sweat information through NFC. (iii) Glucose concentration, pH value, and Cl^- concentration real-time monitoring at 0, 10, and 20 min. (c) A screenshot of the mobile app for the graphical user interface of the SAMS. (d) Glucose concentration, pH value, and Cl^- concentration long-term monitoring during continuous 90 min (30 min rest in between) exercise. (e–g) The consistency of sweat information in different body parts within different time periods. NFC, near field communication; SAMS, sweat-activated system.

sensor develops to a deeper purple. Although the continuous signal data obtained in Figure 4g show that the chloride ion concentration in sweat gradually decreases with exercise time, what is revealed in the colorimetric sensor is the peak concentration in sweat. The color change of the colorimetric sensor validates the practical application of our dual mode sweat monitoring system capabilities.

The detailed electrical information of continuous monitoring when sweating started of glucose concentrations, pH values, and Cl^- concentrations by the SAMS over prolonged physical activity is shown in Figure 4d. Prolonged wear will not cause allergic reactions, as illustrated in Figure S10 in supporting information. After 30 min of exercise followed by a 30 min rest, another 30 min exercise session was conducted to assess sensor

stability and endurance. The subjects took a 30 min break after 30 min of exercise to drink some electrolyte drinks and take in some energy, followed by another 30 min exercise to assess sensor stability and endurance. Another exercise, running, was also tested for 90 min with 30 min rest between to demonstrate the consistency of changes in the three electrical signals, as shown in Figure S9 in supporting information. The curves of glucose concentration, pH value, and chloride ion concentration all decreased gradually during exercise and increased after rest. This is related to the consumption of glucose as an energy source during exercise,^[9,13,70] the pH value is related to the accumulation of lactic acid^[71–73] and the decrease in chloride ion concentration is related to the gradual reduction of electrolytes as sweat is discharged.^[74,75] The observed changes in sweat composition were consistent with expected physiological responses to exercise, highlighting the sensor's precision and reliability.^[9,14,76–78] Throughout this period, real-time sweat data was accessible via a mobile app. In addition, regional and temporal differences in the three sweat components were also studied. Figure 4e–g demonstrates the concentration of glucose and chloride ions and the pH value decreased with the increase of exercise time, and this regional and temporal difference was related to previous reports.^[74,79] Additional data from the other two subjects are provided in Figures S5 and S6 in the supporting information, showing the individual differences.

3 | CONCLUSION

In this study, we employ a multi-mode detection approach that integrates colorimetric and electrochemical techniques to monitor sweat biomarkers such as glucose, pH, and chloride ions. The device is powered by SABs, which offer superior flexibility and biocompatibility, facilitating seamless and safe integration with wearable devices. The continuous secretion of sweat makes SABs well-suited for continuous and long-term monitoring. Additionally, the inclusion of a hydrogel-based colorimetric platform enables the system to provide visual indicators of biomarker levels, enhancing real-time feedback on sweat composition through comparative analysis. Electrochemical detection allows for comprehensive sweat analysis and establishes a robust foundation for continuous health monitoring. Compared to other reported bulky Bluetooth battery systems, our proposed system offers a more convenient, long-term, and power-efficient method for analyzing sweat concentration. It supports diverse monitoring modes, enhancing the efficiency and operation period of

continuous monitoring and detection. Additionally, the NFC technology and mobile phone software design make the system highly practical and promising for long-term sweat monitoring in everyday life. Ultimately, our system achieves a multi-mode, fast-feedback, and long-term sweat monitoring through the integration of colorimetry and electrical signal analysis based on the SABs.

While our proposed system represents a significant advancement in sweat monitoring technology, it is important to recognize and address potential challenges and limitations. Over time, the sensitivity of electrochemical sensors may diminish, potentially impacting the accuracy and reliability of biomarker detection. Additionally, the hydrogel components, essential for the contrasting color platforms, might encounter stability issues under varying environmental conditions, which could affect the system's responsiveness. Data privacy emerges as another critical issue; the incorporation of NFC technology and mobile connectivity necessitates robust security measures to safeguard sensitive health data. To successfully implement and scale this technology, it is crucial to innovate in materials, enhance data security protocols, and develop cost-effective manufacturing processes. Such advancements are essential to meet the diverse needs of users in real-world environments.

4 | EXPERIMENTAL SECTION

4.1 | Chemicals

Phosphate-buffered saline (PBS), chitosan, Nafion, PANI and silver chloranilate were purchased from Sigma Aldrich (St. Louis, MO, USA). Hydroxymethyl EDOT, Chloride ionophore I - cocktail A, PVA (98.0–99.0 mol% Alcoholysis), and graphene were purchased from Aladdin. Manganese dioxide (MnO₂) and sodium chloride (NaCl) were purchased from Macklin (Shanghai, China). Zinc oxide was obtained from Admas-beta. GOx was obtained from BBI. Poly(sodium 4-styrenesulfonate; NaPSS), horseradish peroxidase (HRP), and o-dianisidine dihydrochloride were obtained from Rhawn. Silver/silver chloride (Ag/AgCl) ink was purchased from Ercon Inc. Chlorophenol red was purchased from Shanghai Dibo Biological Technology Co., LTD.

4.2 | PVA-Suc hydrogel

To prepare the PVA-Suc hydrogel, add 1 g of PVA to 5 mL of deionized (DI) water and stirred at 80°C for 3 h. Then, dissolve 4 g of sucrose in 5 mL of DI water while heating

and stirring continuously at 60°C. Next, the sucrose aqueous solution was mixed with the PVA aqueous solution and stirred at 60°C for 3 h to promote thermal cross-linking and hydrogen bond formation between PVA and sucrose. To prepare PVA-Suc-NaCl, mix the PVA solution and sucrose solution with 1 M NaCl, and stir at 60°C for 3 h. Both PVA-Suc and PVA-Suc-NaCl hydrogels can be obtained after cooling to room temperature.

4.2.1 | Electrical sensor

For the fabrication of the electrical glucose sensor, a ZnO solution with a concentration of 10 mg/mL was sprayed onto the sensor substrate at 120°C using a spray gun. After cooling, 2 μ L of GOx at a concentration of 2.5 units/ μ L was pipetted onto the sensor. Following the drying of GOx, 2 μ L of 1% (w/v) chitosan solution was pipetted onto the sensor. To fabricate the electrical pH sensor, PANI films were electrodeposited onto a gold electrode via cyclic voltammetry. The process utilized a voltage window from -0.2 to 1 V, a scan rate of 0.1 V/s, and 60 cycles in a mixture of 0.1 M aniline and 1 M sulfuric acid. A three-electrode system was employed with Ag/AgCl as the reference electrode and Pt as the counter electrode. PANI films grow slowly in the initial stage of polymerization, and the growth rate increases significantly after 20 cycles. In the experiments of preparing PANI films with 50–100 cycles, it was observed that the film thickness of PANI had little effect on the sensitivity of pH, and the hysteresis phenomenon increased slightly with the increase of film thickness.^[80] For the electrical chloride sensor, the working electrode comprised a two-layer structure: a PEDOT layer and a modified ion-selective layer. The PEDOT layer was prepared by immersing an Au electrode in a mixture of 0.01 M EDOT and 0.1 M NaPSS, then depositing PEDOT onto the Au layer using chronopotentiometry with a chi660e electrochemical workstation for 900 s. With a constant current (0.2515 mA) electrodeposition, time determines the thickness of the PEDOT(PSS) film on the Au electrode. The redox process affecting the initial potential is more significant when the PEDOT(PSS) layer on the base electrode is thinner. However, for Au electrodes, the potential differences measured by PEDOT(PSS) layers of different thicknesses are very similar.^[81, 82] The chloride ion-selective membrane was prepared by creating a cocktail solution containing 5.0 wt% 1-decanol, 4.0 wt% 2-nitrophenyl octyl ether, 5.0 wt% chloride ionophore, and 1.0 wt% tetradodecylammonium tetrakis(4-chlorophenyl) borate in 1 mL PBS. 10 μ L of this solution was dropped onto the PEDOT/Au electrode.

4.2.2 | Colorimetric sensor

The colorimetric hydrogel sensor was prepared using the solvent displacement method. Glucose colorimetric reagents were prepared with 40 mL of PBS containing 200 units of HRP, 500 units of glucose oxidase (GOx), and 0.1 mg/mL of o-dianisidine dihydrochloride.^[83, 84] For the pH colorimetric reagents, 0.5% (w/v) chlorophenol red was dissolved in water.^[85] Chloride colorimetric reagents were prepared by mixing 12.5% (w/v) silver chloranilate in water, then dispersing the mixture using a vortex shaker to form a suspension.^[86] For each colorimetric sensor, 40 μ L of the prepared reagents was pipetted onto the hydrogel. The prepared hydrogel was soaked in the reagent for 2 h at room temperature, allowing the reagent to fully penetrate the hydrogel.

4.2.3 | Calibration of the electrical sensor

The electrical signal sensor must be calibrated before the subject wears the SAMS. Calibrate the sensor by soaking it in standard pH6 solution, pH7 solution, DI water, 0.1 mM glucose solution, and 30 mM chloride solution. The measured voltage data is manually input into the mobile app, and the program will automatically calculate an equation corresponding to the potential difference and different ion concentrations. The ambient temperature and the increase in body temperature due to exercise have little effect on the electrical signal sensor, as shown in Figure S11 in the supporting information.

4.2.4 | Sweat-activated batteries

The cathode of the SABs was a magnesium sheet measuring 1 cm in length, 1 cm in width, and 0.1 mm in thickness. The anode is a nickel foam of the same dimensions. A 1:1 mixture of MnO₂ and graphene was added to the nickel foam to catalyze the SAB reaction rates. This mixture was evenly distributed and then dried at 80°C.

ACKNOWLEDGMENTS

X.H., Z.L., X.H., Q.Z., and Y.Z. contributed equally to this work. This work was supported by National Natural Science Foundation of China (Grant no. 62122002), The City University of Hong Kong (Grant nos. 9667221, 9667246, 9680322, and 9667199), Research Grants Council of the Hong Kong Special Administrative Region (Grant nos. RFS2324-1S03, 11211523, 11213721, and 11215722), in part of the InnoHK Project on Project 2.2—

AI-based 3D ultrasound imaging algorithm at Hong Kong Centre for Cerebro-Cardiovascular Health Engineering (COCHE).

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

All human experiments were performed in accordance with protocols approved by the Human Subjects Ethics Sub-Committee of Research Committee, City University of Hong Kong, and conducted in compliance with the guidelines.

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How to cite this article: X. He, Z. Li, X. Huang, Q. Zhang, Y. Zeng, J. Li, C. K. Yiu, Y. Yang, J. Zhou, G. Xu, J. Wang, J. Li, Z. Xu, Z. Chen, Y. Liu, Y. Gao, B. Zhang, G. Zhao, Z. Gao, P. Wu, R. Shi, Y. Qiu, H. Zhang, L. Chow, D. Ye, Y. Huang, X. Yu, *BMEMat* **2025**, 3, e12124. <https://doi.org/10.1002/bmm2.12124>